

Homework 1: Set Theory and Logic

Instructor: Ani Ghosh

Due: Chapter 1, ETC: 1-2 hours, 100 points

Practice Problems: Work together in class

1. Write each of the following sets in set-builder notation.

- (a) The set of all possible quantities q a firm can produce if output must be nonnegative.
- (b) The set of all prices p strictly greater than 0 and less than or equal to 100.
- (c) The set of all income levels m that are at least 50.
- (d) The set of all integer quantities q between 0 and 10, including both 0 and 10.

2. Let

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}, \quad C = \{2, 4\}.$$

Answer the following:

- (a) Find $A \cup B$.
- (b) Find $A \cap B$.
- (c) Is $C \subset A$?
- (d) Is $C \subset B$?
- (e) Is $\emptyset \subset A$?

3. Suppose a consumer can choose bundles (x_1, x_2) of two goods, where both goods must be consumed in nonnegative amounts.

- (a) Write the consumption set as a subset of $\mathbb{R}^2$.
- (b) Is the bundle $(2, 5)$ in the consumption set?
- (c) Is the bundle $(-1, 5)$ in the consumption set?
- (d) Give one economic reason why negative consumption may be ruled out.

4. Let

$$S = \{x \in \mathbb{R} : 10 \leq x \leq 50\}.$$

Interpret S as the set of feasible weekly labor hours for a worker.

- (a) Find an upper bound for S .
- (b) Find a lower bound for S .
- (c) Find the least upper bound of S .
- (d) Find the greatest lower bound of S .

5. For each statement, write the converse and the contrapositive.

- (a) If a firm earns positive profit, then its revenue is greater than its cost.
- (b) If $x > 10$, then $x > 0$.
- (c) If a consumer buys good 1, then the price of good 1 is affordable.

Homework Problems: Submit individually

6. Let

$$D = \{q \in \mathbb{R} : q \geq 0\}$$

be the set of feasible output levels for a firm. Let

$$E = \{q \in \mathbb{R} : q \leq 100\}$$

be the set of output levels that do not exceed the firm's capacity.

- (a) Find $D \cap E$ and interpret it economically.
- (b) Find $D \cup E$.
- (c) If the universal set is $\mathbb{R}$, find D^c .
- (d) Is $D \cap E$ bounded above? If yes, give its least upper bound.

7. Consider the following sets:

$$A = \{x \in \mathbb{R} : x > 5\}, \quad B = \{x \in \mathbb{R} : x \geq 5\}, \quad C = \{x \in \mathbb{R} : x < 20\}.$$

- (a) Find $A \cap C$.
- (b) Find $B \cap C$.
- (c) Are $A \cap C$ and $B \cap C$ the same set?
- (d) Explain the economic difference between “strictly more than 5 units” and “at least 5 units.”

8. Let P be the statement “a consumer's income increases” and let Q be the statement “the consumer's feasible consumption set becomes larger.”

- (a) Write the statement $P \Rightarrow Q$ in words.
- (b) Write the converse of $P \Rightarrow Q$ in words.
- (c) Write the contrapositive of $P \Rightarrow Q$ in words.
- (d) Which is logically equivalent to the original statement: the converse or the contrapositive?

9. Determine whether each statement is true or false. Briefly explain.

- (a) If $A \subset B$, then $A \cup B = B$.
- (b) If $A \subset B$, then $A \cap B = A$.
- (c) If $x \in A \cup B$, then $x \in A$ and $x \in B$.
- (d) If $x \in A \cap B$, then $x \in A$ and $x \in B$.
- (e) If $A = \emptyset$, then $A \subset B$ for any set B .
- (f) The subset order is always a complete order.

10. Suppose an economist claims:

If the price of a normal good falls, then the quantity demanded of that good rises.

- (a) Identify the proposition P and the proposition Q .
- (b) Write the contrapositive of the claim.
- (c) Write the converse of the claim.
- (d) Explain why the converse is not automatically true just because the original claim is true.
- (e) Give one possible reason quantity demanded might rise even if the price of the good did not fall.